

The a priori, possible worlds, de re modality, and origin essentialism

PHIL UA-78 Metaphysics

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Necessity and a priority

The concept of a priori knowability

Some philosophers writing in the earlier part of the twentieth century seem to equate claims of metaphysical necessity with claims about *a priori knowability*—things that can be known “without relying on sense-experience”.

- ▶ The question what it even *means* to know something ‘without relying on sense-experience’, and whether it’s even possible to know anything in this way, is an interesting one, but one for epistemology rather than metaphysics—we are only mentioning the idea because we want to set it aside.

A basic point from Lecture 1 of N&N is that these are different notions: the claim that something is metaphysically necessary has nothing to do with knowledge or thought.

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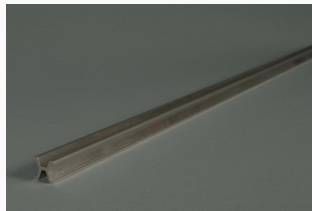
A less basic point is that either of ‘It is metaphysically necessary that P ’ and ‘It is knowable a priori that P ’ can be true without the other being true!

The necessary a posteriori and contingent a priori

- ▶ It is metaphysically necessary that Hesperus is Phosphorus TRUE (by NI)
- ▶ It is knowable a priori that Hesperus is Phosphorus SEEMS FALSE
- ▶ It is knowable a priori that if anything is one metre long, the standard metre is one metre long SEEMS TRUE
- ▶ It is metaphysically necessary that if anything is one metre long, the standard metre is one metre long SEEMS FALSE



Venus. Patrick Myers, US National Park Service (public domain).



US National Prototype Metre no. 27. NIST Research Library & Museum (public domain).

The a priori of identity?

Worry: Last week we gave an argument for the claim that it is necessary that Hesperus is Phosphorus. What will happen when we re-run this argument, substituting 'it is knowable a priori that' everywhere where we had 'it is necessary that'?

Will we be forced to say that it is, in fact, knowable a priori that Hesperus is Phosphorus, that being golden is being composed of atoms with 79 protons, etc.?

The a priority of identity?

Well, let's consider. The argument went in three main stages:

1. NSI: For any object x , it is necessary that $x = x$ (premise).
2. NI: For any object x and object y , if $x = y$, it is necessary that $x = y$ (by 1 and LL).
3. It is necessary that Hesperus = Phosphorus (from 2 and 'Hesperus = Phosphorus').

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So the argument we need to consider will have corresponding stages:

1. NSI*: For any object x , it is knowable a priori that $x = x$ (premise).
2. NI*: For any object x and object y , if $x = y$, it is knowable a priori that $x = y$ (by 1 and LL).
3. It is knowable a priori that Hesperus = Phosphorus (from 2 and 'Hesperus = Phosphorus').

We *might* worry about premise 1. Maybe there are some objects (particles? points? ghosts?) which no-one could know a priori to be self-identical, because no-one could even think about them at all.

But, we don't need the full strength of NSI to run this argument—it would be enough if we said 'for every planet in the solar system' rather than 'for every object'. And it seems clear that each of those *can* be known a priori to be self-identical.

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The best place to push back is in the step from 2 to 3.

- 'Know' is an *attitude verb*, and takes part in the same strange behavior we noted last week for 'want'. The truth or falsity of a sentence of the form 'X knows that *P*' turns not only on *what state of affairs* we refer to with '*P*', but on *what form of words* we use to refer to it. And this is also true for 'X knows a priori that *P*', 'Someone could know a priori that *P*', etc.

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- ▶ By contrast, the truth or falsity of 'It is metaphysically necessary that *N* is *F*' depends just on what object we refer to with '*N*' and what property we refer to with '*F*'.

Epistemic modals

Many modal words in English have an *epistemic* use which can be roughly paraphrased with 'know'.

- ▶ 'Class might be cancelled today' $\approx$ 'For all I know, class is cancelled today'
- ▶ 'There must be something wrong with the heater' $\approx$ 'We know (based on this salient evidence) that there is something wrong with the heater'.

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On the epistemic reading, it seems like 'Samuel Clemens must be Mark Twain' could be false even if 'Samuel Clemens is Mark Twain' is true. Does that mean that if Kripke is right that 'It is metaphysically necessary that Samuel Clemens is Mark Twain' is true, we have to give up on our characterization of metaphysical necessity as the 'broadest' necessity?

I suggest that the answer is *no*. Epistemic ‘must’ is like ‘know’, sensitive to what words are used as well as to what objects/properties/states of affairs they refer to.

So, when we say ‘It is not the case that Clemens must be Twain’ or ‘It is not epistemically necessary that Clemens is Twain’ we are not saying that this state of affairs—that Clemens is Twain, i.e. that Clemens is Clemens—lacks a certain status we are calling ‘epistemic necessity’.

What we are saying is compatible with the claim that this state of affairs is necessary in every sense.

Possible worlds

Possible worlds talk

Thanks largely to Kripke, analytic philosophers love to express claims about necessity and possibility using the expression 'possible world', initially introduced by Leibniz.

They assume the **Leibniz biconditionals**:

It is possible for it to be the case that P if and only if at some possible world, P .

It is necessary that P if and only if at every possible world, P .

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It is necessary that P if and only if at every possible world, P .

P if and only if at the actual world, P .

This turns out to be a useful way of talking—very helpful for tracking the logical relationships between modal claims!

- In his earlier work, Kripke pioneered something called *possible worlds model theory* inspired by the Leibniz biconditionals, which had a revolutionary impact on modal logic.

Note that this is not limited to *metaphysical* possibility and necessity: one can treat 'It is physically necessary that P ' as equivalent to 'At every physically possible world, P ', and so on.

Possible worlds as parallel universes?

It is easy to slip into talking about possible worlds as if they were other planets, or parallel universes just as real as the universe we live in.

If you do this, you'll think of 'At some possible world, there are talking donkeys' as implying that there *are* talking donkeys out there in reality (though maybe not in the small chunk of reality we call 'the actual world').

David Lewis, in his book *On the Plurality of Worlds*, actually defends this view. The resulting metaphysical theory is fascinating, but most philosophers find it completely unbelievable. Most are confident that it is metaphysically possible for there to be talking donkeys, but also confident that there just aren't any.

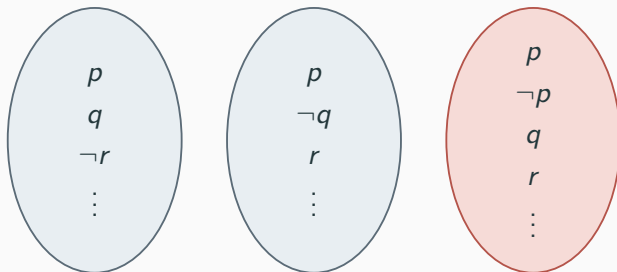
In this course, we won't be discussing this theory of Lewis's.

What *are* possible worlds, if not parallel universes?

Happily, rejecting Lewis's ambitious theory doesn't have to mean giving up on possible worlds talk. There are less "exciting" interpretations of what it is to be a possible world that make the Leibniz biconditionals plausible.

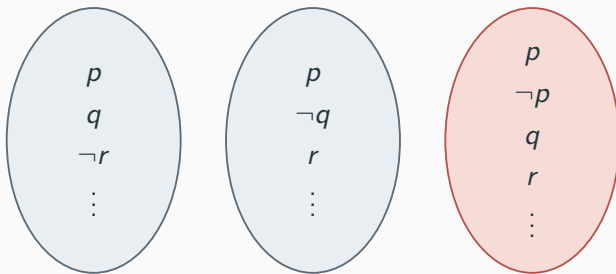
Here's my favorite approach:

- A world is a collection of states of affairs that contains every state of affairs or its negation.



What *are* possible worlds, if not parallel universes?

- ▶ 'At world w , P ' means that the state of affairs that P is a member of the collection w .
- ▶ ' w is actual' means 'every state of affairs in w obtains'.
- ▶ ' w is possible' means 'it is possible that w is actual'. (So every interpretation of 'possible' gives an interpretation of 'possible world'.)



What *are* possible worlds, if not parallel universes?

On this approach, the Leibniz biconditionals amount to the claim that if a state of affairs is possible, we can always build it up to a collection that contains every state of affairs or its negation where it's possible that every member of that state of affairs obtains. This seems at least a bit intuitive, though certainly not obvious.

A note on how I'm using 'state of affairs'

We've been talking freely about *properties*, equating 'object x is F ' with 'object x has the property of being F ' (at least when we don't have weird words like 'want' and 'know' floating around).

This practice extends to *binary relations*—we can equate 'Antony loves Cleopatra' with 'The *loving* relation holds of Antony and Cleopatra (in that order)'. And similarly for ternary, quaternary, . . . relations.

Properties can be equated with 1-ary relations. They hold, or fail to hold, of just one object.

States of affairs can similarly be equated with 0-ary relations. They just hold or fail to hold *full stop*—we don't need to feed them any arguments before we can ask 'true or false'.

When we have a 2-ary relation like *loving* and an object, we can ‘plug’ the object in to one or other argument place to make a property: *loving Cleopatra*, *being loved by Cleopatra*.

In exactly the same way, we can ‘plug’ an object (Cleopatra) into a property (*being Egyptian*) to make a state of affairs (*that Cleopatra is Egyptian*).

- These are not the only ways to ‘build’ $n - 1$ -ary relations out of n -ary relations. E.g. we can also use *existential quantification*, which can turn *love* into either *loving something* or *being loved by something*, and can turn *being Egyptian* into the state of affairs *that something is Egyptian*.

A note on 'proposition'

Some authors (including me) use 'proposition' as a synonym for what I am calling 'states of affairs'.

But others tie 'proposition' more tightly to the mysteries of attitude verbs. For example, they might say that the *proposition* that Hesperus is bright is not identical to the proposition that Phosphorus is bright, and that's why someone could believe the former and not the latter. They give 'the proposition that...' the same semantic peculiarity as 'believes', 'wants', 'knows', etc.

By contrast, we assume that since Hesperus = Phosphorus, the property *living on Hesperus* is identical to the property *living on Phosphorus*, and by the same token the state of affairs *that Hesperus is bright* is identical to the state of affairs *that Phosphorus is bright*.